bSQUARE
The Mobile & Embedded Systems Experts

Hobart Corporation



HLX Scale

Software Architecture Document

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Submitted by:

BSQUARE Corporation
110 110th Ave NE Suite 200
Bellevue, WA 98004
(425) 519-5900 (Voice)
(425) 519-5999 (Fax)

To:

Hobart Corporation
701 S. Ridge Ave
Troy, OH 45374

Attn: Debra Dyer
Telephone #

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Project Name	HLX Scale Linux Port
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Author(s)	Minhaj Ahmed Sr. Software Engineer II Bsquare Professional Engineering Services (PES) 2100 Lakeside Blvd. St. 450 Richardson, TX 75080
Reviewed by	Michael Browne Director of Engineering 2100 Lakeside Blvd , Suite 450 Richardson, TX 75082
Approved by:	Bsquare Project Manager: John Imboden Sr. Program Manager 3480 West Market Street, suite 105 Fairlawn, OH 44333 Client Project Manager: Name Debra Dyer Title Division
Delivery Date	

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Contents

DEFINITIONS, ACRONYMS, AND ABBREVIATION 4

APPLICABLE DOCUMENTS 4

OVERVIEW 5

SUPER USER ACCESS 5

Definitions, Acronyms, and Abbreviation

The following terms and acronyms are used throughout this document.

Bsquare	Bsquare Corporation
sudo	Ubuntu command used to execute in super user mode.
OS	Operating System
root	Administrator's account in Linux/Unix operating systems.

Applicable Documents

The following table contains documents that were referenced during the creation of this document.

Ref. #	Document Name	Document ID/Location
1	Linux Platform Setup.doc	https://externalsvn.camelot.bsquare.com/svn/Hobart/Hobart_Linux_Port/branches/shared_code/Docs/Tech docs
2	HLX OS Paths.xlsx	https://externalsvn.camelot.bsquare.com/svn/Hobart/Hobart_Linux_Port/branches/shared_code/Docs/Tech docs

Table 1 – Applicable Documents

Overview

HLX platforms that are setup in accordance with the *Linux Platform Setup* document already provide appropriate access to directories and files for a typical user. This document covers general methodology adopted by Ubuntu in order to be able to perform essential administrative tasks.

Super user access

The Administrator account in the Linux operating system (OS) is known as *root*. Like other OSs *root* account has full control of the system and is able to make any necessary changes to the system. For this reason most Linux OSs discourage running in Administrator mode and implement various schemes to be able to appropriately manage the system when required. On platforms running Ubuntu the initial user created during installation is given majority of the administrative rights. However, in order to properly guard administrative functionality, the *sudo* command is required for commands executed on the command-line. This scheme allows the users to be able to execute specified command as an administrator without logging out of the system. Since password is required before the execution of the command, users not having access to the password are unable to perform administrative tasks. For example:

```
sudo mkdir /etc/hobart
```

would allow an administrator to create *hobart* directory in an otherwise restricted */etc* directory. The sudo command would cause the user to be prompted for password and the operation would only succeed if a correct password is entered. Operation would simply fail if the sudo command was not specified at all. Graphics based utilities implement similar behavior by locking critical administrative tasks and password is required to enable them. Additionally, in order to further inhibit access to utilities which may change system behavior, all menu items are removed from the desktop during the setup process limiting the user to Hobart application specific user interface.